Chapter 3
Derivatives

MAE101 - CALCULUS

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The Tangent Problem

1 Introduction

- Find an equation of the tangent line to the y = f(x) at the point $P(a, f(a))$.
- The slope of the secant line PQ:

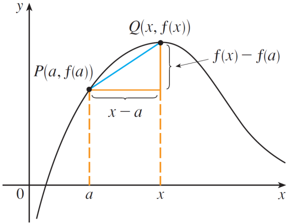
$$m_{PQ} = \frac{f(x) - f(a)}{x - a}$$

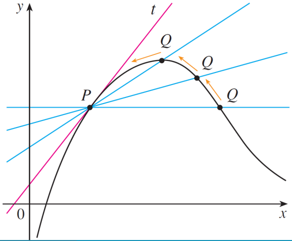
- The slope of the tangent line:

$$m = \lim_{Q \to P} m_{PQ} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

- The equation of the tangent line:

$$y = m(x - a) + f(a)$$





Tangent lines

1 Introduction

Denifition

The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.

Example 1. Find an equation of the tangent line to the parabola $y = x^2$ at $P(1, 1)$.

Example 2. Find an equation of the tangent line to the hyperbola $y = 3/x$ at $(3, 1)$.

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2 Derivatives

Derivatives

2 Derivatives

Definition

The derivative of a function f at a number a, denoted by f′(a), is

$$f'(a) = \lim_{x \to \mathrm{a}} \frac{\mathrm{f(x) - f(a)}}{\mathrm{x - a}}$$

if this limit exists.

or

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

If we replace a in above formular by a variable x, we obtain

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

OTHER NOTATIONS.

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x); \quad \left.\frac{dy}{dx}\right|_{x=a} \text{ or } \left.\frac{dy}{dx}\right]_{x=a}$$

Derivatives - Example

1. Show that a constant function $f(x) = k$ has derivative $f'(x) = 0$.
2. Find $f'(x)$ if $f(x) = sinx$
3. Find $f'(0)$ if $f(x) = \begin{cases} \dfrac{\sqrt{1+2x}-1}{x} & , x \neq 0 \\ 1 & , x = 0 \end{cases}$
4. Find $f'(0)$ if $f(x) = \sqrt[3]{x}$

Differentiable and continuous

2 Derivatives

Definition

A function f is **differentiable at a** if f′(a) exists.
It is **differentiable on an open interval** (a, b) [or (a, ∞) or $(-\infty$, a) or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

Theorem

f is differentiable at $a \Longrightarrow f$ is continuous at a.

Note:

f is differentiable at $a \not\Leftarrow f$ is continuous at a.

f is discontinuous at $a \Longrightarrow f$ is NOT differentiable at a.

Example. $f(x) = |x|$ is continuous at $x = 0$, but NOT differentiable at $x = 0$.

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3 Formulas

Differentiation formulas

1. $\frac{d}{dx}(k) = 0.$
2. $\frac{d}{dx}(x^n) = nx^{n-1}, n \in \mathbb{R}.$
3. $\frac{d}{dx}\sin x = \cos x.$
4. $\frac{d}{dx}\cos x = -\sin x.$
5. $\frac{d}{dx}\sec x = \sec x \tan x.$
6. $\frac{d}{dx}\tan x = \sec^2 x.$
6. $\frac{d}{dx}\csc x = -\csc x \cot x.$
7. $\frac{d}{dx}\cot x = -\csc^2 x.$
8. $\frac{d}{dx}(e^x) = e^x.$
9. $\frac{d}{dx}(lnx) = \frac{1}{x}.$
10. $\frac{d}{dx}(a^x) = a^x lna.$

Differentiation rules

3 Formulas

Name of rule	Function notation
Constant multiple	$[cf(x)]' = cf'(x)$
Sum and difference rule	$[f(x) \pm g(x)]' = f'(x) \pm g'(x)$
Linearity rule	$[af(x) + bg(x)]' = af'(x) + bg'(x)$
Product rule	$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$
Quotient rule ($g(x) \neq 0$)	$\left[\dfrac{f(x)}{g(x)}\right]' = \dfrac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$

Example. Differentiate the functions:

1. $f(x) = \sin x + \cos x$
2. $g(t) = t^2 + \cos t + \cos \dfrac{\pi}{4}$
3. $f(x) = 2x^3 \sin x - 3x \cos x$
4. $p(x) = x^2 \cos x.$
5. $q(x) = \dfrac{\sin x}{x}.$

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4 Higher-Oder

Higher-Order Derivatives

			Leibniz notation
First derivative:	y'	$f'(x)$	$\dfrac{dy}{dx}$ or $\dfrac{d}{dx}f(x)$
Second derivative	y''	$f''(x)$	$\dfrac{d}{dx}\left(\dfrac{dy}{dx}\right)=\dfrac{d^2y}{dx^2}$ or $\dfrac{d^2}{dx^2}f(x)$
Third derivative	y'''	$f'''(x)$	$\dfrac{d}{dx}\left(\dfrac{d^2y}{dx^2}\right)=\dfrac{d^3y}{dx^3}$ or $\dfrac{d^3}{dx}f(x)$
$\vdots$	$\vdots$	$\vdots$	
n^{th} derivative	$y^{(n)}$	$f^{(n)}(x)$	$\dfrac{d^ny}{dx^n}$ or $\dfrac{d^n}{dx^n}f(x)$

Example. Find f', f'', f''', and $f^{(4)}$.

1. $f(x) = x^5 - 5x^3 + x + 12.$
2. $f(x) = \dfrac{1}{4}x^8 - \dfrac{1}{2}x^6 - x^2 + 2.$

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Chain rule

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F′ is given by the product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if y = f(u) and u = g(x) are both differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Example 1. Find the Derivative of the functions: a. $y = \sin 4x$ b. $y = \sqrt{4 + 3x}$.

Example 2. Let $f(x) = g(sin3x)$. Find f' in terms of g'.

Example 3. Suppose $h(x) = f(g(x))$ and $f(2) = 3$, $g(2) = 1$, $g'(2) = -1$, $f'(2) = 2$, $f'(1) = 5$. Find $h'(2)$.

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6 Implicit

To find the derivative of the implicit form:

- **Step 1:** Differentiate both sides of the equation with respect to x. Remember that y is really a function of x for part of the curve and use the chain rule.
- **Step 2:** Solve the differentiated equation algebraically for $\frac{dy}{dx}$

Example. Find $\frac{dy}{dx}$ by implicit differentiation

1. $x^2y + 2y^3 = 3x + 2y$
2. $x^2 + y^2 = 25.$
3. $x^2 + y = x^3 + y^3.$
4. $xy(2x + 3y) = 2.$
5. $\frac{1}{y} + \frac{1}{x} = 1.$
6. $\tan\frac{x}{y} = y.$
7. $\cos xy = 1 - x^2.$
8. $e^{xy} + 1 = x^2.$
9. $e^{xy} + \ln y^2 = x.$
10. $\ln(xy) = e^{2x}.$

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7 Logarithmic

Logarithmic Differentiation

Logarithmic differentiation is especially valuable as a means for handling complicated product or quotient functions and exponential functions where variables appear in both the base and the exponent.

Example.

1. Find $\dfrac{dy}{dx}$, where $y = (x+1)^{2x}$.
2. Find the derivative of $y = \dfrac{e^{2x}(2x-1)^6}{(x^3+5)^2(4-7x)}$ if $y > 0$

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8 Applications

Rate of change

The derivative can be interpreted as a rate of change, which leads to a wide variety of applications. Viewed as rates of change, derivatives may represent such quantities as the speed of a moving object, the rate at which a population grows, a manufacturer's marginal cost, the rate of inflation, or the rate at which natural resources are being depleted.

- Average rate of change $= \dfrac{\text{CHANGE IN } y}{\text{CHANGE IN } x} = \dfrac{\Delta y}{\Delta x} = \dfrac{f(x+\Delta x)-f(x)}{\Delta x}$
- Instaneous rate change $= \lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x} = \lim\limits_{\Delta x \to 0} \dfrac{f(x+\Delta x)-f(x)}{\Delta x} = f'(x)$
- Relative rate of change $= \dfrac{f'(x)}{f(x)}$

Example. Let $f(x) = x^2 - 4x + 7$

1. Find the average rate of change of f with respect to x between $x = 3$ and $x = 5$.
2. Find the instantaneous rate of change of f at $x = 3$.

An object that moves along a straight line with position $s(t)$ has

1. Velocity $v(t) = \dfrac{ds}{dt}$
2. Acceleration $a(t) = \dfrac{dv}{dt} = \dfrac{d^2s}{dt^2}$
3. The speed of the object is $|v(t)|$

when these derivatives exist.

Example. Assume that the position at time t of an object moving along a line is given by

$$s(t) = 3t^3 - 40.5t^2 + 162t$$

for t on $[0, 8]$. Find the initial position, velocity and acceleration for the object and discuss the motion. Compute the total distance traveled.

FORMULA FOR THE HEIGHT OF A PROJECTILE

$$h(t) = -\frac{1}{2}gt^2 + v_0t + s_0$$

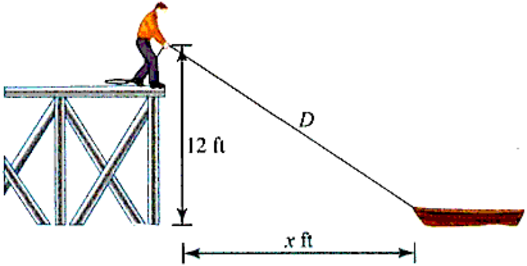
- v_0 is the initial velocity.
- s_0 is the initial height.
- g is the acceleration due to gravity ($32ft/s^2$ or $9.8m/s^2$)

Example.

Suppose a person standing at the top of the Tower of Pisa (176ft high) throws a ball directly upward with an initial speed of 96ft/s. Find the ball's height, its velocity, and its acceleration at time t.

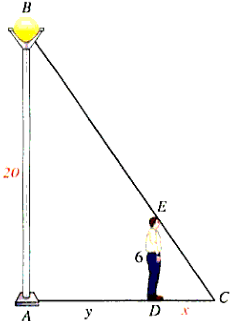
SOLVING RELATED RATE PROBLEMS

- **Step 1** Draw a figure, if appropriate, and assign variables to the quantities that vary. Be careful not to label a quantity with a number unless it never changes in the problem.
- **Step 2** Find a formula or equation that relates the variables. Eliminate unnecessary variables.
- **Step 3** Differentiate the equations. You will usually differentiate implicitly with respect to time.
- **Step 4** Substitute specific numerical values and solve algebraically for any required rate.



A person is standing at the end of a pier 12ft above the water and is pulling in a rope attached to a rowboat at the waterline at the rate of 6ft of rope per minute, as shown in Figure. How fast is the boat moving in the water when it is 16ft from the pier?

Related rates and Applications - Example



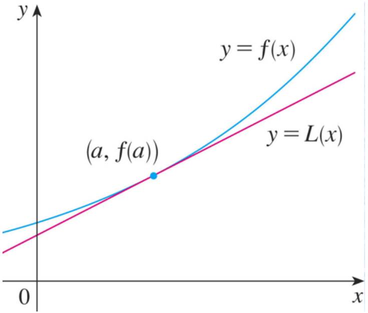
A person 6ft tall is walking away from a street light 20ft high at the rate of 7ft/s. At what rate is the length of the person's shadow increasing?

We use the tangent line at $(a, f(a))$ as an approximation to the curve y = f(x) when **x is near** a

$$L(x) = y = f(a) + f'(a)(x - a)$$

The approximation $f(x) \approx L(x)$ is called the linear approximation of f at a.

$$|x - a| \leq \epsilon, \forall \epsilon \geq 0$$



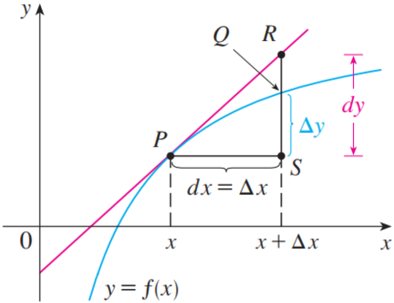
Example. Determine the linear approximation for $f(x) = \sqrt[3]{x}$ at $x = 8$. Use the linear approximation to approximate the value of $\sqrt[3]{8.05}$ and $\sqrt[3]{25}$.

Differential

Given a function $y = f(x)$ we call dy and dx differentials and the relationship between them is given by

$$dy = f'(x)dx$$

Differentials provide us with a way of estimating the amount a function changes as a result of a small change in input values.



Example 1. Compute the differential for $y = t^3 - 4t^2 + 7t$.

Example 2. The radius of a sphere was measured and found to be 21 cm with a possible error in measurement of at most 0.05 cm. What is the maximum error in using this value of the radius to compute the volume of the sphere? ($dV = 4\pi r^2 dr$)

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Problems

1. Find an equation of the tangent line to the curve at the given point

   a. $y = \dfrac{x-1}{x-2}$, $(3,2)$ b. $y = \dfrac{2x}{x^2+1}$, $(0,0)$ c. $y = 3 - 2x + x^2$, $x = 1$

2. Find all the values of x where the tangent line to the graph of the function is horizontal $f(x) = x^3 + 4x^2 - 11x + 11$.

3. Find the derivative of the following function

   a. $y = \dfrac{x}{\sin x}$ b. $y = xe^{-2x}$ c. $y = \dfrac{e^{x^2}}{\sin x}$

4. Find y''

   a. $y = xe^{3x-1}$ b. $y = \sqrt[3]{2x+1}$ c. $y = e^{-x}\cos x$

5. The position in feet of race car along a straight track after t seconds is modeled by the function $s(t) = 8t^2 - \dfrac{1}{16}t^3$

   a. Find the average velocity of the vehicle over the time interval $[4, 4.1]$

   b. Find the instantaneous velocity of the vehicle at $t = 4$ seconds.

6. A toy company can sell x electronic gaming systems at a price of $p = -0.01x + 400$ dollars per gaming system. The cost of manufacturing x systems is given by $C(x) = 100x + 10000$ dollars. Find the rate of change of profit when 10000 game produced. Should the toy company increase or decrease production? (profit=$x.p - c(x)$).
7. A table of values for f, f', g and g' is given

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	2	4	6
2	1	8	5	7
3	7	2	7	9

   a. If $h(x) = f(g(x))$, find $h'(1)$ b. If $H(x) = g(f(x))$, find $H'(1)$
8. If $h(x) = \sqrt{4 + 3f(x)}$, where $f(1) = 7, f'(1) = 4$, find $h'(1)$.

Problems

9. Find f' in terms of g': a. $f(x) = g(\sin 2x)$ b. $f(x) = g\left(e^{1-3x}\right)$.
10. Find $\dfrac{dy}{dt}$ for:

    a. $y = x^3 + x + 2, \dfrac{dx}{dt} = 2$ and $x = 1$. b. $y = \ln x, \dfrac{dx}{dt} = 1$ and $x = e^2$
11. Find y' by implicit differentiation

    a. $x^4 + y^4 = 16x + y$ b. $x^3 + xy = y^2$
12. Find the linearization $L(x)$ of the function at a

    a. $f(x) = \dfrac{1}{\sqrt{2+x}}$, $a = 2$ b. $f(x) = \sqrt[3]{5-x}$, $a = -3$.
13. Each side of a square is increasing at a rate of 7 cm/s. At what rate is the area of the square increasing when the area of the square is 25 cm^2?
14. Compute the differential of the function $y = e^{2x^2}$.

Q&A

Thank you for listening!